

Security and Protection of Systems

Lecture 04 *Firewalls*

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Outlines

- Firewall
- Virtual Private Network
- Mail Gateway
- Intrusion Detection System
- Intrusion Prevention System

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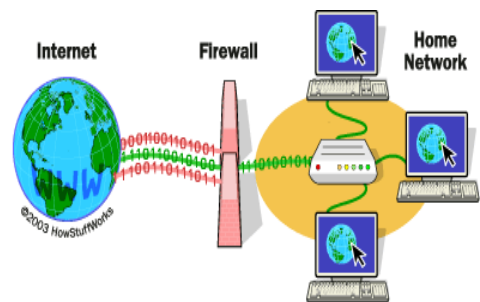
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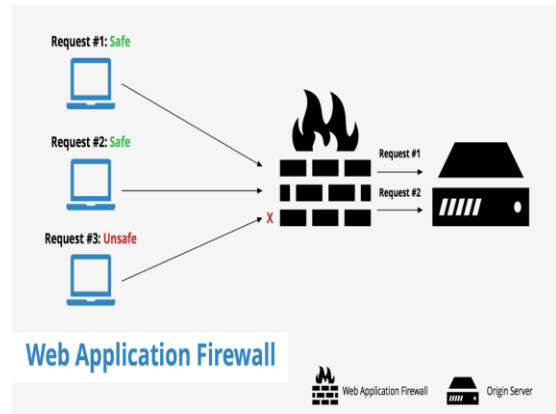
Firewalls

- A firewall is a kind of [cybersecurity tool](#) used to control network traffic.
- Firewalls may be used to [isolate network nodes](#), [internal sources](#), or [even special programs](#) from external traffic sources.
- The main objective of a firewall is to [block malicious network requests](#) and data packets while allowing legitimate traffic
- Firewalls can be classified as [Web Application](#) - [Network Segmentation](#) - [Database](#) - [Cloud](#) - [Next Generation](#).



Web Application-Based Firewalls (1)

- Is a [proxy server](#) between server application and users that accesses from outside the network.
- Takes input and then creates a connection [on behalf](#) of the internal client with the request.



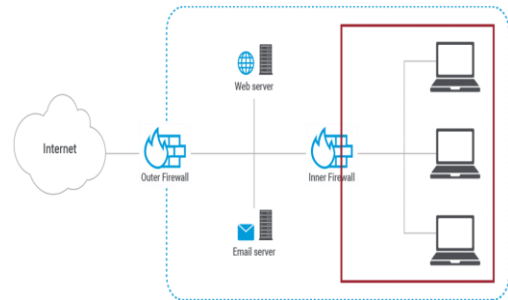
Web Application-Based Firewalls (1)

- **Level of Protection:** High because the web application server offers a buffer for unidentified and potentially malicious users who could otherwise have direct access to the Web application server.
- **Strengths:** Web application firewalls are simpler, less vulnerable, and easier to patch than web servers themselves.
 - This means that hackers can consider applications behind the firewall substantially difficult.
- **Weaknesses:** proxy firewalls do not support all applications easily and can reduce the safe application performance for end-users.

Network Segmentation-Based Firewalls (2)

- An **internal** network firewalls
- Used to **manage** network traffic flows between locations, operational areas, divisions, or other business units.
- It is applied at **subnet limits**, can be a network breach in one area and not throughout the network.

Multiple Network Security Perimeters



Network Segmentation Firewalls (2)

- **Level of Protection:**
- It can also protect areas of the network that it guarantees, such as databases or research and development units.
- An attacker may be unable to move from part of a network to another
- Can only slow attacker progress in practice if the initial break is quick to identify.
- **Strengths:** For very big companies or companies with network perimeters that are difficult to secure, network segmentation firewalls are most helpful.
- **Weaknesses:** If an aggressor achieves network access, then it can be significantly more difficult for a network segmentation firewall to access particularly sensitive information.

Database Firewalls (3)

- A type of firewall for Web applications designed to protect databases.
- Usually installed right onto the server of the database .
- They aim to identify and avoid unique server attacks, such as cross-site scripts, which can lead to confidential information in databases accessed by attackers.



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Database Firewalls (3)

Level of Protection:

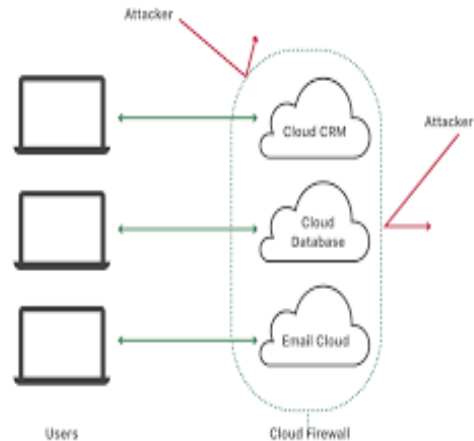
- The loss of confidential information is usually expensive and costly regarding lost credibility and poor ads.
- When hackers continue to target databases effectively, this means that records are increasingly important.
- **Strengths:** Can provide an effective security measure and can also be used to track, review and report compliance for regulatory purposes.
- **Weaknesses:** Only if configured and modified correctly and offer little protection from zero-day exploits will they be effective.

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Cloud-based Firewalls (4)

- A cloud-based firewall is an alternative to a corporate data center firewall but has the same aim: to protect a network, application, database, or other IT resources.

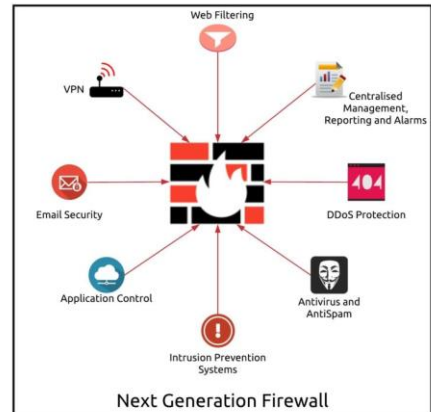


Cloud-based Firewalls (4)

- **Level of Protection:** The security professional who specializes in firewall management configures and manages a cloud firewall as a service to offer excellent protection for the resources.
- It will also be highly accessible with little or no scheduled or unplanned downtime.
- It is usually done with company routers' configuration to divert traffic to the Cloud firewall when mobile users connect to it either via a VPN or as a proxy.
- **Strengths:** The configuring of a container firewall is probably easier than a host firewall that operates on every container.
- **Weaknesses:** Could be wasteful and difficult to justify on a cost basis in smaller settings.

Next-generation Firewalls (5)

- Used to protect network from undesired data traffic, but they are distinct from conventional firewalls.
- In addition to its port, origin, destination IP address, and protocol, NGFWs provide software visibility with full-Stack Visibility by looking at each data packet's contents.
- It allows you to prohibit the use of specific applications such as peers for file-sharing applications in application layers and limit applications .
- Provides better network firewall coverage than a conventional firewall, leaving costs and performance problems on the one hand.



Next-generation Firewalls (4)

- **Level Protection:** Quite high because they provide a high degree of granular control.
- **Strengths:** NGFW has far more grain control of data, enabling the NGFW to address a broader range of potential threats and can't access the corporate network
- **Weaknesses:** NGFWs cost more than traditional firewalls, which can cause network performance problems because they conduct packet inspection rather than just packet filters.

Outlines

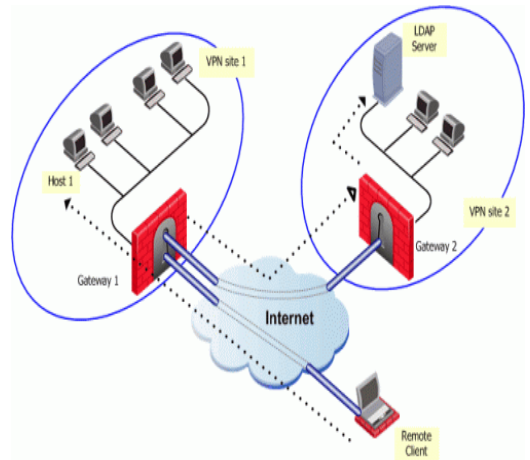
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Virtual Private Network (VPN)

- VPN enables authorized users to use an unsecured public network as if it were a secure private network
- All data transmitted between remote device and network is encrypted
- Types of VPNs
 1. Remote-access VPN - a user-to-LAN connection
 2. Site-to-site - multiple sites can connect to other sites over the Internet

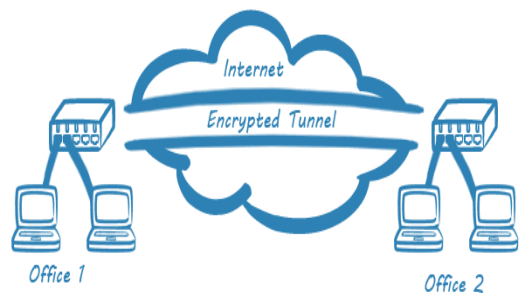
Remote Access VPN

- Permits a user to connect to a private network and access all its services and resources remotely.
- The connection between the user and the private network occurs through the Internet and the connection is secure and private.

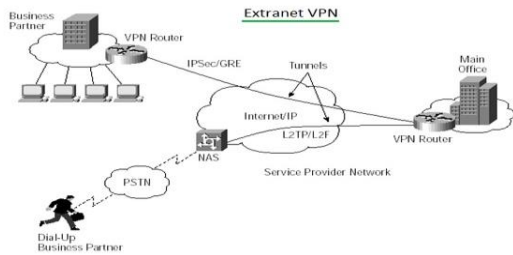


Site to Site VPN

- Also called as Router-to-Router VPN
- Organizations, with different branches, use to connect the network.
- One router acts as a VPN Client and another as a VPN Server.
- Create a imaginary bridge between networks via Internet.
- When authentication is validated between the two routers only then the communication starts.



Site to Site VPN



1. **Intranet based VPN:** When several offices of the **same** company are connected using Site-to-Site VPN type, it is called as Intranet based VPN.

2. **Extranet based VPN:** When companies use Site-to-site VPN type to connect to the office of **another** company, it is called as Extranet based VPN.

VPN Protocols (1) (IPSec)

1- Internet Protocol Security (IPSec):

- Used to secure Internet communication across an IP network.
- Secures Internet Protocol communication by verifying the session and encrypts each data packet during the connection.
- Runs in 2 modes: (i) **Transport mode** and (ii) **Tunneling mode**
- The work of transport mode is to encrypt the message in the data packet and the tunneling mode encrypts the whole data packet.
- IPSec can be used with other security protocols to improve the security system.

VPN Protocols (1) (IPSec)

Pros

1. **Network Security:** operates at network level; completely invisible operations, and traffic are fully monitored.
2. **Application In-dependency:** There is no application dependency here.
3. **Data Privacy:** safeguarded using public keys.
4. **Network Support:** Due to the presence of IP Layer; IPSec can be implemented to any network from all the sizes.
5. **Authentication:** signatures on each packets.

Cons

1. **CPU Overhead:** Transmitted data needs to be encrypted and decrypted constantly.
 - Requires high processing power from the CPU.
2. **Algorithms:** Weak Security algorithms.
 - Greater level of risk.
3. **Access Range:** IPSec provides wide access range,
 - Higher chances of giving privileges to other devices in the network.

VPN Protocols (2) (L2TP)

2. Layer 2 Tunneling Protocol (L2TP):

- L2TP or Layer 2 Tunneling Protocol is a tunneling protocol that is often combined with another VPN security protocol like IPSec to establish a highly secure VPN connection.
- L2TP generates a tunnel between two L2TP connection points and IPSec protocol encrypts the data and maintains secure communication between the tunnel.

VPN Protocols (2) (L2TP)

Pros

1. **Security:** Extremely secure; doesn't contain any major vulnerabilities and Uses 256 bits encryption key.
2. **Compatibility:** built onto and supported by different operating systems.
3. **Configurations:** Easy configuration and setup; most platforms natively support.
4. **Stability:** When establishing connection, L2TP is considered to be very stable and reliable.

Cons

1. **Speed:** encapsulating data twice can be degraded speed.
2. **Security Keys Mismatch:** the Pre-shared keys require alternation at both ends of the tunnel. If there is a mismatch between them, the L2TP will stop working.
3. **Port Support:** supports limited number of ports. If it is used behind firewall they can be easily blocked because they are limited in numbers.
4. **Eavesdropping:** uses Pre-shared keys can be taken over by the provider's website.

VPN Protocols (3) (PPTP)

3- Point-to-Point Tunneling Protocol (PPTP):

- PPTP generates a tunnel and confines the data packet.
- PPTP is used to encrypt data between the connection.
- PPTP is one of the most widely used VPN protocol and has been in use since the early release of Windows, and also used on Mac and Linux apart from Windows.

VPN Protocols (3) (PPTP)

Pros.

Cons

- | | |
|--|---|
| <ol style="list-style-type: none"> 1. Compatibility: Compatible with all operating Systems 2. Usability: seamless setting up since there is no data encryption involved. 3. Cost: cheap, and the installation process don't need to be an expert. 4. Internet Speed: establishing a VPN connection gradually does not reduce Internet speed. | <ol style="list-style-type: none"> 1. Performance: faces performance issues when it is used on unstable networks. 2. Security: low; limited to 128 bit encryption only. 3. Reliability: less reliable since it lacks data origin verification and data integrity process. 4. Firewall Restrictions: Internet Service Providers (ISPs) often impose firewalls that recognizes and blocks PPTP protocols. |
|--|---|

VPN Protocols (4) (SSL and TLS)

4- SSL and TLS:

- SSL (Secure Sockets Layer) and TLS (Transport Layer Security) generate a VPN connection where the web browser acts as the client and user access is prohibited to specific applications instead of entire network.
- Online shopping websites commonly uses SSL and TLS protocol.
- It is easy to switch to SSL by web browsers and with almost no action required from the user as web browsers come integrated with SSL and TLS.
- SSL connections have "https" in the initial of the URL instead of "http".

VPN Protocols (4) (SSL and TLS)

Pros

1. **Data Integrity:** certificate being exchanged is specified by the TLS; ensures that no any transmitted data gets lost.
2. **Trustworthy:** reliable website confident enough to customers to conduct transactions.
3. **Security:** it HMAC (Key-Mashing Authentication Code).; method ensures that transmitted record cannot be tampered

Cons

1. **High Latency:** secure encryption methods provides higher latency.
2. **MiM Attacks:** some versions of TLS is still vulnerable to Man in the Middle Attacks.
3. **Platform Support:** some versions such as TLS 1.3 that is only able to support few of the platforms.
4. **Implementation Cost:** Implementing a TLS certificate isn't free, there are some costs involved.
5. **Network Complexity:** Complex network.

VPN Protocols(5) (OpenVPN)

5- OpenVPN:

- OpenVPN is an open source VPN that is commonly used for creating Point-to-Point and Site-to-Site connections.
- It uses a traditional security protocol based on SSL and TLS protocol.

Pros

1. Security levels are the highest -
2. Reliable for its users
3. The support of community all over the world

Cons

1. There are certain proxy servers that are not compatible and do not support OpenVPN.
2. Sometimes the latency can be high.
3. Not really user friendly

VPN Protocols (6) (SSH)

6- Secure Shell (SSH):

- Secure Shell or SSH generates the **VPN tunnel** through which the data transfer occurs and also ensures that the tunnel is **encrypted**.
- SSH connections are generated by a SSH client and data is transferred from a local port on to the remote server through the encrypted tunnel.

VPN Protocols (6) (SSH)

Pros

1. Proven technology: **robust and reliable**.
2. Strong international encryption.
3. Both free and commercial versions exist.
4. SSH client runs on most platforms, the server runs on **UNIX, Linux** and **VMS**.
5. Tunnelling of static TCP ports works well and can be automated to use for simple VPNs.
6. Many authentication methods

Cons

1. Port ranges & dynamic ports can't be forwarded.
2. SSH server daemon: **Cannot restrict what ports may or may not be forwarded, per user**.
3. Performance can be a problem on old machines.
4. Port forwarding can also introduce security problems.

Comparison	VPN	Proxy
Security	Offers high amount of security; ensures encryption, authentication and integrity protection.	Proxy does not ensure or provide any security.
Protocol Being Use	PPTP and L2TP	FTP , SMTP, HTTP
Environment	Works on firewall.	Works on browsers.
Simulation	Simulate a private network over public network.	Does not simulate a private network over public network.
IP Address	Does not hide the IP address of client.	Uses the anonymous network ID instead of actual IP address of client.
Tunnel Creation	VPN creates tunnel between end users.	Does not create tunnel between end users.

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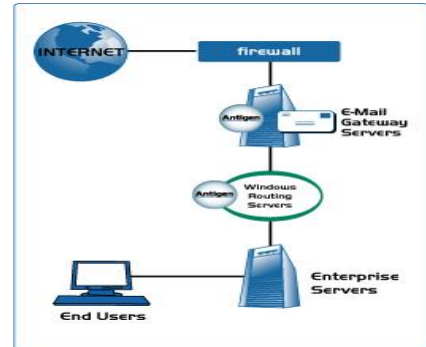
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Mail Gateway

- The **mail gateway** is a machine that handles connections between networks that run different communications protocols or communications between different networks that use the same protocol.
- For example, a mail gateway might connect a TCP/IP network to a network that runs the Systems Network Architecture (SNA) protocol suite.
- Monitors emails for unwanted content and prevents these messages from being delivered
 1. For inbound emails, *can search for various types of Malware, spam, and phishing attacks*
 2. For outbound email, *can detect and block the transmission of sensitive data*



Mail Gateway

Things to consider when selecting a secure email gateway are:

1. **Deployment Options:** *whether an organization wants to deploy email security as a cloud service or on-premises.*
2. **Spam Effectiveness:** *organizations should review email archiving solution comparisons.*
3. **Threat Intelligence:** *the ability to detect and block malware, threat intelligence can help administrators understand how they are being attacked,*
4. **Outbound Content Control:** *Analyzing email leaving an organization to prevent sensitive data from leaving or automatically using encryption to enforce security policy.*
5. **Response Capabilities:** *If an email makes it to a user inbox and is subsequently found to be malicious, the ability to automatically locate and remove those emails will prevent subsequent infection.*

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Intrusion Detection System (IDS)

- **IDS** is a system that monitors **network traffic** for suspicious activity and issues alerts when such activity is discovered.
- It is a **software application** that scans a network or a system for the **harmful activity or policy breaching**.
- Any malicious **venture** or **violation** is normally reported either to an administrator or collected centrally using a security information and event management (**SIEM**) system.
- A **SIEM** system integrates outputs from multiple sources and uses alarm filtering techniques to differentiate malicious activity from false alarms.

Intrusion Detection System (IDS)

Detection Methods

1. Anomaly-based monitoring: Compares current detected behavior with baseline
2. Signature-based monitoring: Looks for well-known attack signature patterns
3. Behavior-based monitoring : Detects abnormal actions by processes or programs and Alerts user who decides whether to allow or block activity
4. Heuristic monitoring: Uses experience-based techniques

Intrusion Detection System (IDS)- Type

1. Host intrusion detection system (HIDS)
 - A software-based application that can detect an attack as it occurs
 - Installed on each system needing protection
 - Monitors:
 - System calls and file system access
 - Can recognize unauthorized Registry modification
 - Host input and output communications
 - Disadvantages
 - Cannot monitor network traffic that does not reach local system
 - All log data is stored locally
 - Resource-intensive and can slow system

Intrusion Detection System (IDS)- Type

2. Network intrusion detection system (NIDS)

- Watches for attacks on the network
- NIDS sensors installed on firewalls and routers:
- NIDS can sound an alarm and log events
- Application-aware IDS
 - A specialized IDS
 - Uses "contextual knowledge" in real time
 - It can know the version of the OS or which application is running
 - As well as what vulnerabilities are present in the systems being protected

Intrusion Detection System (IDS)- Type

3- Protocol-based Intrusion Detection System (PIDS):

- PIDS comprises a system or agent that would consistently resides at the **front end of a server**, **controlling and interpreting** the protocol between a user/device and the server.
- It is trying to secure the web server by regularly **monitoring** the HTTPS protocol stream and accept the related HTTP protocol.
- As HTTPS is **un-encrypted** and before instantly entering its web presentation layer then this system would need to reside in this interface, between to use the HTTPS.

Intrusion Detection System (IDS)- Type

4- Application Protocol-based Intrusion Detection System (APIDS):

- APIDS is a system or agent that generally resides within a group of servers.
- It identifies the intrusions by monitoring and interpreting the communication on application-specific protocols.
- .For example, this would monitor the SQL protocol explicit to the middleware as it transacts with the database in the web server.

Intrusion Detection System (IDS)- Type

5- Hybrid Intrusion Detection System :

- Hybrid intrusion detection system is made by the combination of two or more approaches of the intrusion detection system.
- In the hybrid intrusion detection system, host agent or system data is combined with network information to develop a complete view of the network system.
- Hybrid intrusion detection system is more effective in comparison to the other intrusion detection system. Prelude is an example of Hybrid IDS.

Comparison of IDS with Firewalls:

- IDS and firewall both are related to network security but an IDS differs from a firewall as a firewall looks outwardly for intrusions in order to stop them from happening.
- Firewalls restrict access between networks to prevent intrusion and if an attack is from inside the network it doesn't signal.
- An IDS describes a suspected intrusion once it has happened and then signals an alarm.

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Intrusion Prevention Systems (IPSs)

- It is a network security application that **monitors** network or system **activities** for **malicious activity**.
- Major functions of IPSs are to **identify malicious activity**, **collect information** about this activity, **report it** and **attempt to block or stop it**.
- IPSs are contemplated as augmentation of IDS because both IPS and IDS operate network traffic and system activities for malicious activity.
- IPS typically record information related to observed events, notify security administrators of important observed events and produce reports.

Intrusion Prevention Systems (IPSs) Types

1. **Network-based intrusion prevention system (NIPS):** It **monitors the entire network** for suspicious traffic by analyzing protocol activity.
2. **Wireless intrusion prevention system (WIPS):** It **monitors a wireless network** for suspicious traffic by analyzing wireless networking protocols.
3. **Network behavior analysis (NBA):** It **examines network traffic** to identify threats that generate unusual traffic flows, such as distributed denial of service attacks, specific forms of malware and policy violations.
4. **Host-based intrusion prevention system (HIPS):** It is an inbuilt software package which operates a single host for doubtful activity by scanning events that occur within that host.

IPS Technology Type	Types of Malicious Activity Detected	Scope per Sensor	Strengths
Network-Based	Network, transport, and application TCP/IP layer activity	Multiple network subnets and groups of hosts	Only IDPS which can analyze the widest range of application protocols;
Wireless-Based	Wireless protocol activity; unauthorized WLAN in use	Multiple WLANs and groups of wireless clients	Only IDPS able to predict wireless protocol activity
NBA-Based	Network, transport, and application TCP/IP layer activity that causes anomalous network flows	Multiple network subnets and groups of hosts	more effective than the others at Identifying DoS attacks, and reconstructing malware infections
Host-Based	Host application and OS activity; network, transport, and application TCP/IP layer activity	Individual host	Can analyze activity that was transferred in end-to-end encrypted communications

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(IPSs) Detection Methods

- 1. Signature-based detection:** operates packets in the network and compares with pre-built and preordained attack patterns known as signatures.
- 2. Statistical anomaly-based detection:** monitors network traffic and compares it against an established baseline.
- 3. Stateful protocol analysis detection:** recognizes divergence of protocols stated by comparing observed events with pre-built profiles of generally accepted definitions of not harmful activity.

Comparison of IPS with IDS:

1. IPS are **placed in-line** and are able to actively prevent or block intrusions that are detected.
2. IPS can take such actions as sending an **alarm**, **dropping** detected malicious packets, **resetting** a connection or blocking traffic from the offending IP address.
3. IPS also can **correct cyclic redundancy check (CRC)** errors, defragment packet streams, mitigate TCP sequencing issues and clean up unwanted transport and network layer options

Thanks